CSET485 – AI and Society: Assignment #1

Roll number: s24bcau0169 | **Dataset rule:** last digit 9 → Adult Census Income | **Random seed:** 169 (70% train / 30% test)

Target: income > $50K encoded as 1 (yes) / 0 (no). **Notebook:** CSET485_Assignment1.ipynb (fully reproducible; every number below regenerated by running it end-to-end).

Data Preprocessing (Documented)

Missing values. The raw file encodes unknowns as **"?"**: workclass 1,836; occupation 1,843; native-country 583. These were treated as missing and the affected rows were **dropped** (2,399 rows removed from 32,561 raw rows), leaving 30,162 records.

Target encoding. income → 1 if >$50K else 0. Positive rate = 24.89%.

Categorical encoding. One-hot encoding (drop='first') on workclass, education, marital-status, occupation, relationship, race, sex, native-country (8 columns → 90 binary features). education-num kept numeric.

Feature scaling. StandardScaler (zero mean, unit variance) on continuous predictors age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week. One-hot columns left as 0/1.

Outliers. Skewed continuous features winsorized at the 1st/99th percentile; bounds computed on the **training** split only and then applied to the test split (no leakage). Train winsorization (below 1st / above 99th): age 0/83, fnlwgt 59/96, education-num 65/0, capital-gain 0/72, capital-loss 0/85, hours-per-week 80/51.

Split (seed 169, stratified): train 21113 (positive rate 0.2489); test 9049 (positive rate 0.2489). Final feature matrix: 96 columns (6 scaled continuous + 90 one-hot).

Part 1 — Build & Compare Model A vs Model B

Model A = Ordinary Least Squares (Linear Probability Model) on the binary income target. **Model B** = Logistic Regression predicting P(income > $50K).

Model A (OLS): R^2 = 0.3984, median predicted value = 0.2297, test accuracy = 0.7183.

Model B (Logistic): test accuracy = 0.8520, AUC = 0.9068, intercept = -3.0970.

Confusion matrix of agreement/disagreement (rows = Model A, cols = Model B):

	Model B: high risk	Model B: low risk
Model A: high risk	1881	2644
Model A: low risk	0	4524

Mismatches: **2644 / 9049 = 29.22%**. All mismatches are Model A “high risk” vs Model B “low risk” (2644 cases); 0 cases the other way.

Why they differ. Model A thresholds at its own median predicted value (**0.2297**), far below the logistic **0.50** probability cutoff. Because the OLS median is much lower than 0.5, Model A labels many more people “high risk.” One rule splits a continuous score at its median; the other splits calibrated probabilities at 0.5 — with a low median, they flag different individuals.

Part 2 — The Trap

Sensitive variable: race (Group A = White; Group B = Black).

2.1 Simpson’s Paradox hunt

Predictor: **education-num** (logistic regression, 95% CI).

Model	n (train)	Coefficient	95% CI	p-value
Full	21,113	0.8262	[0.7838, 0.8686	0.00e+00
Group A (White)	18,124	0.8124	[0.7676, 0.8573	1.0e-275
Group B (Black)	2,012	1.0310	[0.8396, 1.2224	4.7e-26

No sign reversal, but the return to education is **markedly stronger for Black** workers (coef 1.0310 vs 0.8124, ~27% larger). Each added year of schooling raises the log-odds of earning >$50K more for Black workers. **Policy implication if ignored:** a blanket "more education → higher income" conclusion masks that the payoff differs by race, so investment could be misallocated and the benefit to Black workers under-estimated.

2.2 Omitted-Variable Bias

Dropped confounder: education (categorical). Before/after coefficients (full model vs model without education):

Predictor	Full coef	Without education	% change
education-num	1.031205	0.696564	-32.45%
age	0.355970	0.368319	+3.47%
capital-gain	0.860972	0.859616	-0.16%
hours-per-week	0.385786	0.389075	+0.85%

education-num shifted the most (-32.45%). education-num was proxying for the dropped education categories; removing education cut its coefficient by a third. A model can look stable while a key coefficient is badly distorted — coefficients are not trustworthy until confounds are stress-tested.

2.3 Adversarial Subset

High-confidence wrong predictions on the test set: **455 / 9049 = 5.03%** (FP: 107 with P≥0.75 but actual ≤$50K; FN: 348 with P≤0.25 but actual >$50K). A 50-row sample (seed 169) is saved in adversarial_subset.csv.

Subset vs full test set: older (mean age 43.5 vs 38.3), more male (70% vs 67.5%), more White (90% vs 86.3%). **Standout characteristic: 62% are age ≥ 40 vs 43.5% in the full test set** (also 88% work ≥40 hrs/week vs 78%).

Why the model fails confidently. These are older, high-hours workers whom the linear logistic boundary rates very unlikely (P≤0.25) yet who actually earn >$50K (false negatives), or rates very likely yet who do not (false positives). The boundary is linear in scaled features and was learned from a younger, higher-capital-gain population; missing age × capital-gain interactions / non-linear effects cause confident failure on this predictable subgroup.

2.4 Fairness Trade-Off Analysis

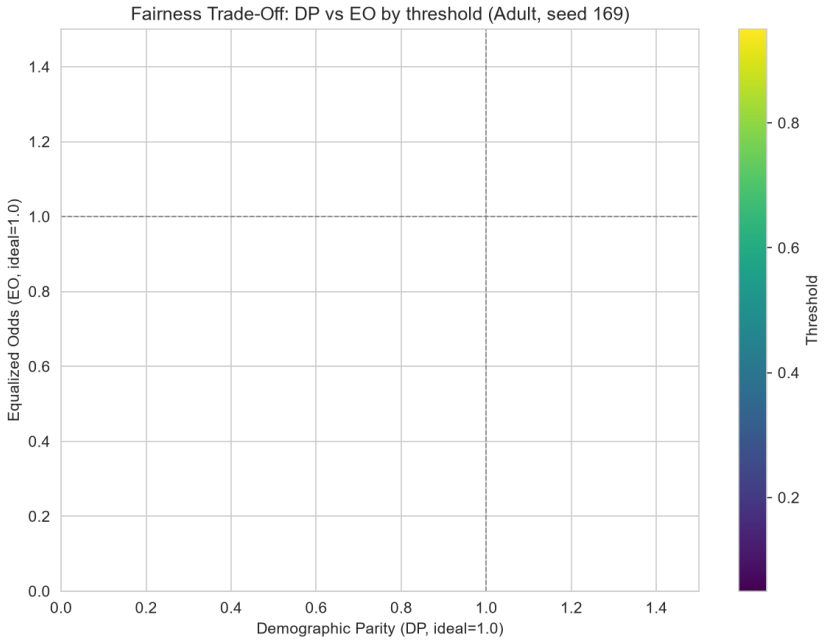
Sensitive: race (White vs Black). Logistic threshold swept 0.05 → 0.95 (step 0.1). DP = P(pred=1|White)/P(pred=1|Black); EO = min(TPR ratio, FPR ratio); PP = P(Y=1|pred=1,White)/P(Y=1|pred=1,Black). Ideal = 1.0.

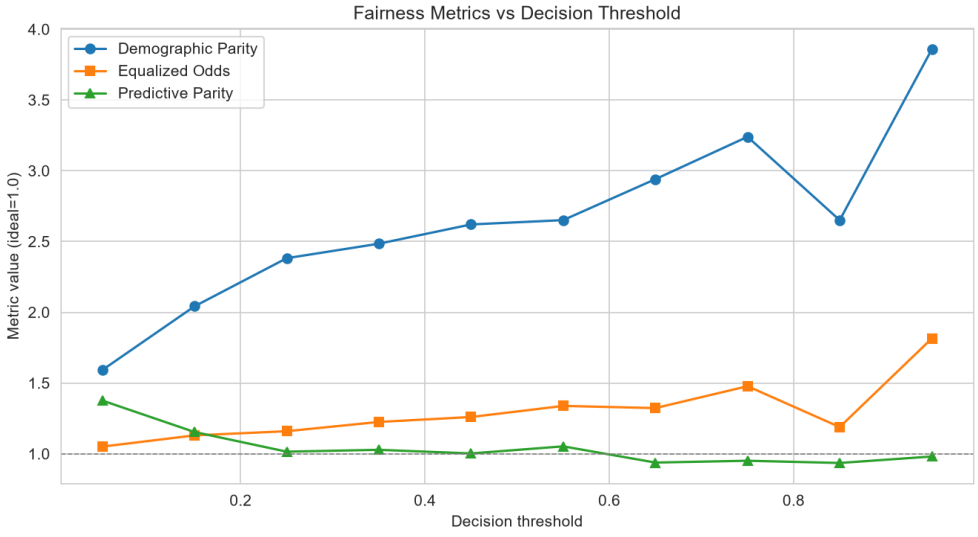
Threshold	DP	EO	PP	Accuracy
0.05	1.592	1.051	1.376	0.631
0.15	2.042	1.130	1.154	0.758
0.25	2.382	1.160	1.015	0.811
0.35	2.484	1.225	1.028	0.838
0.45	2.621	1.260	1.002	0.849
0.55	2.651	1.338	1.053	0.851
0.65	2.940	1.323	0.938	0.845
0.75	3.240	1.478	0.951	0.832

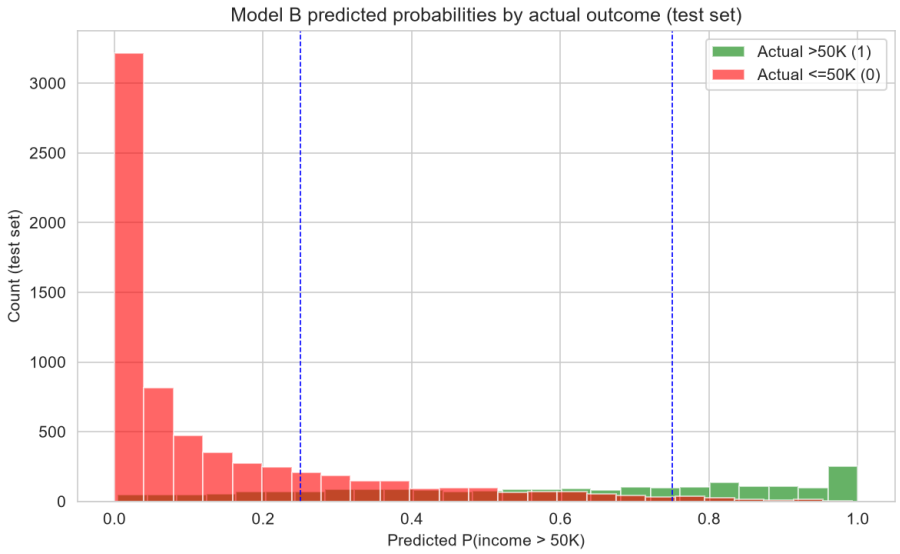
Threshold	DP	EO	PP	Accuracy
0.85	2.651	1.189	0.935	0.808
0.95	3.858	1.815	0.981	0.781

Trade-off. DP and EO both move toward their ideal as the threshold lowers (DP 1.592 at t=0.05 vs 3.858 at t=0.95). Lowering the threshold to improve DP (e.g. 2.621→1.592 from t=0.45 to t=0.05, a ~63% reduction in DP disparity) comes at the cost of predictive parity (PP 1.002→1.376) and accuracy (0.849→0.631). No single threshold makes DP, EO and PP all equal 1.0.

Recommended threshold: t ≈ 0.45 — PP = 1.002 (near-perfect predictive parity), accuracy = 0.849, with DP = 2.621 / EO = 1.260. This sacrifices demographic parity (White predicted positive ~2.6× Black) to give both groups similar conditional accuracy among those predicted positive.






Model B predicted probabilities by actual outcome (test set)
Actual >50K (1)
Actual <=50K (0)
Count (test set)
0
500
1000
1500
2000
2500
3000
0.0
0.2
0.4
0.6
0.8
1.0
Predicted P(income > 50K)

Part 3 — Descriptive Analysis Questions

Q1 — Threshold-based vs direct logistic classification

Model A (OLS) thresholds at its median predicted value of **0.2297**; Model B thresholds at **0.50**. On the 9,049-row test set the rules agreed high/high on 1,881 and low/low on 4,524, but **mismatched on 2,644 cases (29.22%)** — all Model A "high risk" vs Model B "low risk" (zero cases the other way). Because the OLS median score (0.2297) sits far below the logistic 0.5 probability cutoff, Model A flags roughly a quarter of the test set as high risk that Model B does not. The approaches flag different individuals because one splits a continuous score at its median while the other splits calibrated probabilities at 0.5; with a low median the OLS rule is far more permissive, so the same person can be "high risk" under Model A and "low risk" under Model B.

Q2 — Subgroup / Simpson's Paradox

Split by race (White n=18,124; Black n=2,012 in training). Predictor = education-num. Full model: **coef 0.8262 [0.7838, 0.8686]**. White: **0.8124 [0.7676, 0.8573]**. Black: **1.0310 [0.8396, 1.2224]**. No sign reversal, but the education return is ~27% larger for Black workers (point estimate 1.0310 vs 0.8124). Ignoring the subgroup produces the blanket policy "more education → higher income," underestimating the payoff for Black workers and overstating uniformity; targeted support would be misallocated and the Black subgroup's greater return to schooling missed.

Q3 — Omitted-variable bias (dropped education)

Predictor	Full coef	Without education	% change
education-num	1.031205	0.696564	–32.45%
age	0.355970	0.368319	+3.47%
capital-gain	0.860972	0.859616	–0.16%
hours-per-week	0.385786	0.389075	+0.85%

education-num shifted the most (−32.45%). It was proxying for the dropped categorical education variable. Without stress-testing for confounds, a key coefficient can be off by a third, so coefficient magnitudes are not trustworthy until confounds are checked — a model can look stable while a central predictor is badly distorted.

Q4 — Detecting an unseen hidden confound

Commission an external validation on a different, temporally-later census sample (or the CPS ASEC) with the same features plus a rich set of domain variables (e.g., local unemployment, occupation prestige, health status), then compare subgroup performance and calibration. Any systematic miscalibration or performance gap in the external data that the current metrics hide signals a missing confound (e.g., a local labor-market factor) not captured by the original features. Residual analysis by subgroup would corroborate it.

Q5 — The 50-row adversarial subset

455 test rows (5.03%) were high-confidence wrong; the sampled 50 are older (mean 43.5 vs 38.3), more often White (90% vs 86.3%), and **62% are age ≥ 40 vs 43.5% of the full test set** (also 88% work ≥40 hrs vs 78%). The model's linear decision boundary underestimates older, high-hours workers with modest capital-gain: they truly earn >$50K but receive P≤0.25 (false negatives), because age × capital-gain interactions and non-linear effects are absent. The model extrapolates from a younger, wealthier training pattern and fails confidently on this predictable subgroup.

Q6 — Trade-off cost for a real person

Compare t=0.45 (DP=2.621, EO=1.260, PP=1.002) vs t=0.05 (DP=1.592, EO=1.051, PP=1.376). Two applicants, one White and one Black, each truly earning >$50K. At t=0.45 both have similar odds of approval (predictive parity ≈1). At t=0.05 approval rates equalize more across groups, but a positive decision no longer means the same thing for both (PP=1.376): an approved White candidate is less likely to truly qualify. Prioritizing demographic parity helps the Black

applicant get approved more often, but harms the White applicant by weakening the meaning of the decision — fairness to one group is bought with weaker conditional accuracy for the other.

Q7 — Defensible threshold (t = 0.35)

At t=0.35: DP=2.484, EO=1.225, PP=1.028, accuracy=0.838. PP≈1.03 (near-perfect predictive parity: approved White and Black applicants are about equally likely to truly qualify), accuracy is high (83.8%), and EO=1.23 (modest TPR/FPR gap). It is defensible because it equalizes the meaning of a positive decision across groups while keeping overall error low. Counterargument: DP=2.484 still means White applicants are ~2.5× as likely to be approved as Black applicants — a critic would say this entrenches group-level access inequality even if conditional accuracy is fair. That is valid: PP fairness does not fix unequal selection rates.

Q8 — Demographic parity vs equalized odds

I recommend **equalized odds** for this context. My numbers: at t=0.45, EO=1.260 vs DP=2.621 — EO is much closer to parity (1.0) than DP, while PP=1.002 (near-perfect predictive parity) and accuracy=0.849. Equalized odds keeps true-positive and false-positive rates similar across groups, so qualified applicants of either race face similar chances — what matters most when the decision affects livelihood. I sacrifice demographic parity (approval rates remain ~2.6× higher for White applicants). That is acceptable because equalizing rates without equalizing accuracy would approve many unqualified applicants from the disadvantaged group (harmful stigma and default risk), while equalized odds treats both groups by the same performance standard.

Part 4 — Reflection

Direct answer: I would recommend **conditional deployment, not deployment-as-is**: the model is accurate overall (AUC 0.907, accuracy 85.2%) but fails confidently on a predictable subgroup (5.03% high-confidence errors; 62% of the adversarial sample aged ≥40), so it should be deployed only with threshold t≈0.45, subgroup monitoring, and a human-appeal path.

Why a single number fails. (1) Overall accuracy hides subgroup gaps: test accuracy is 85.2%, yet the model is confidently wrong on 5.03% of cases, and those errors concentrate on older workers (62% aged ≥40 vs 43.5% of the test set) — a single accuracy figure conceals this. (2) Threshold choice is a values question, not a data-driven fact: moving from t=0.05 to t=0.95 changes DP from 1.592 to 3.858 and accuracy from 0.631 to 0.781 — the “best” threshold depends on how much disparity versus accuracy you will tolerate. (3) Fairness metrics conflict: DP and EO both improve as the threshold drops, but predictive parity worsens (PP 1.376 at t=0.05 vs 1.002 at t=0.45); no single threshold makes DP, EO and PP all equal 1.0. (4) The model confidently fails on a predictable subset: older, high-hours workers with modest capital-gain receive P≤0.25 yet actually earn >$50K; accuracy alone never reveals this. (5) Hidden confounds remain unknowable: dropping education moved education-num’s coefficient by −32.45%, so an unseen confound could distort the model just as badly without any metric flagging it.

Grounding in evidence. Accuracy 85.2% / AUC 0.907 (Part 1); threshold mismatch 29.22% (Part 1); education-num −32.45% shift when education is dropped (Part 2.2); 455 high-confidence errors = 5.03%, 62% aged ≥40 (Part 2.3); no threshold makes DP/EO/PP all equal 1.0 (Part 2.4).

Closing. Responsible deployment requires publishing the full trade-off surface — threshold, subgroup error rates, and the fairness metrics in tension — rather than asking a policymaker for a yes/no on one number.